

3 Building a Baseline for Quantitative Risk Assessments

We conducted a risk assessment to better understand the threat of criminal action against an in-house system. We examined both cybercrimes and physical crimes using data that was normalized to a per-capita rates of crimes. We drew from FBI Internet Crime Complaint Center (IC3) annual reports and Chattanooga Police Open Data records. The population samples were taken from the US Census Bureau reports. We were able to build a baseline simulation that can be used to better understand the context of the likelihood and severity of observed events in the lab.

3.1 Introduction

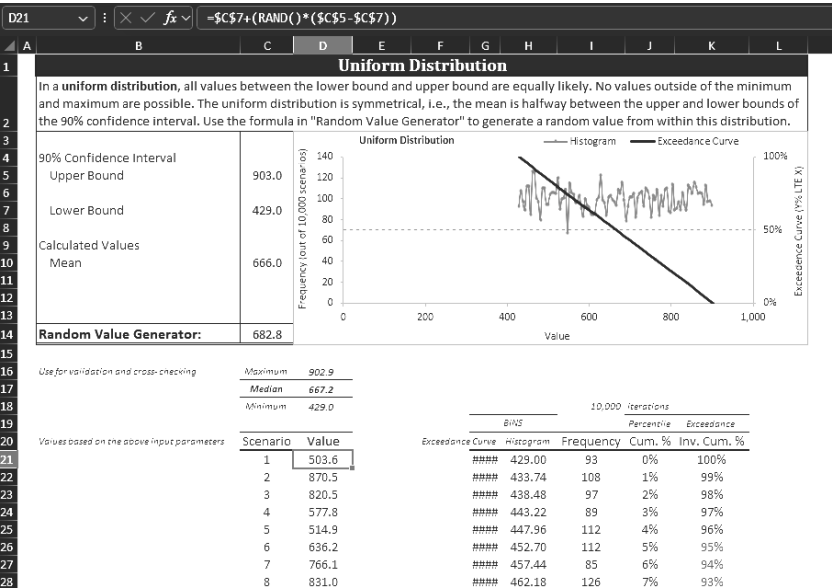


Fig. 3.1: Example run of a Monte Carlo simulation using state-level data to set bounds. Here we chose Personal Data Breach victim counts from 2016 and 2022 as upper and lower bounds. 10,000 rows projected randomized values between these bounds and determined mean and counts by confidence interval. [1]

3.1.1 Background and Content

We had a modest amount of education in statistics as we began the project. Our skills were derived primarily from three college courses. At Chattanooga State Community College in 2011, we had courses in "Elementary Statistics." [2] This was an algebra-based course that focused on the fundamentals of applying data to normal

distributions. Also there in 2012, we had a course in "Statistics for Engineers." [3] This was an algebra-based course that discussed various quality control techniques and statistical modeling. In 2023 at Harvard University Extension School, we had a graduate course in "Risk in Information Security" with Derek Brink. [4] Professor Brink authored the Excel spreadsheet which we have used for Monte Carlo simulations.

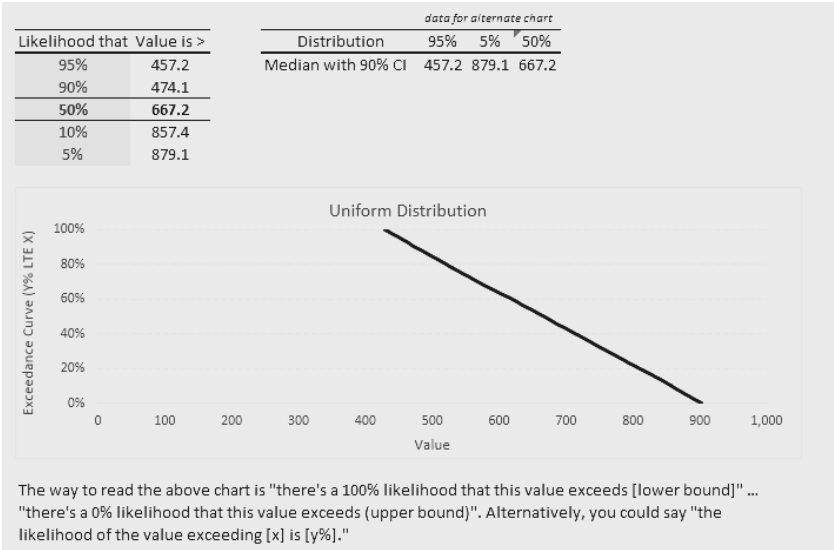


Fig. 3.2: Example results of applying Monte Carlo simulation to forecasting. Here we see the assertion that there is a 90% likelihood that the number of victims will be between 457 and 879; the mean of the simulation was 667.[5]

The spreadsheet conducts 10,000 simulations of data points between user-supplied bounds. Sheets in the Excel workbook are capable of conducting a variety of these simulations. They are differentiated by modeling style (uniform, triangular, normal, binary, or discrete) and points of interest (count, cost, or percentage).

Each of the simulating rows follows these general steps:

- Reference the lower bound.
- Ascertain the range between lower and upper bounds.
- Generate a random number.
- Multiply the random number against the range value.
- Install the generated random value in a cell.

The row values are then counted, sorted, and ranked into confidence intervals. The values and critical points are plotted on various graphs. The end result is that it is easy for an operator of the spreadsheet to generate a Monte Carlo simulation of a distribution between bounds. Since there are enough iterations (10,000), the simulation can have statistical value. Since the inputs on the spreadsheet can be readily modified and since the cells are calculated upon page saving or loading, new sim-

ulations can be run with ease. We recognized Professor Brink's spreadsheet was a useful tool that we could apply to forecasts of interest to us here.

3.1.2 Problem Statement

We need a cheap source of risk information to inform procedural decisions in the lab. We can't afford to pay a vendor for intelligence feeds. Free attack-specific feeds that work well for defense technologies like Snort or Suricata are not in a form that would inform policy-level risk decisions. We want to gather this information efficiently and without generating any civil or criminal liability or threats; we're not going to conduct firsthand research in cybercrime forums to determine what shape a localized risk might take. We want to be able to use publicly available cybercrime data in order to pursue commercial-grade decisions on a localized scale in a manner that is informed and practical.

These conditions suggest that we should use the following resources:

- United States (US) IC3 data
- Computer Emergency Response Team (CERT) data from other nations
- Open Data from local government agencies
- MITRE ATT&CK for guiding data selection
- Other scientifically valid Open Data sources.

We found during our research that there would sometimes be promising documents published by security vendors and software companies. Unfortunately, many of those were not accompanied by scientifically valid open data sources. So it is with caution that we could sometimes accept the assertions; however, we sometimes found that the reports were useless beyond identifying an anecdotal cause, condition, or effect. Without data to quantify those identified conditions, we were sometimes left with a good idea that could not be proven in a manner that supported a quantitative risk assessment. In those times, those leads had to sometimes be abandoned or be allowed to fall short of their potential impact. Unscientific assertions from vendors seemed to be a trend in our research environment; unscientific assertions from vendors is such a part of the problem's environment that it is a problem almost unto itself.

3.1.3 Project Objectives

We want to be able to conduct a quantitative risk assessment for our small project lab that would be scientific and illustrative enough to allow others to accept it as guidance.

We want others to be able to conduct their own risk assessments and be able to use these notes as a pattern for their own benefit.

We want to be able to use scaled-to-size IC3 data in a Monte Carlo simulation as a baseline for understanding the potential rates and severity of cybercrime encounters.

We want to proceed with an educated, professionally supported philosophical framework of ideas. We didn't want to decay into philosophical edge cases like

harshly questioning cause and effect, blindly ignoring unquantified threats, or emotionally fueling conspiratorial fears. We need to keep our feet on the ground and our systems defended; but, if the risk assessment shows that threats are rare or severe, then we need to understand that through quantitative analysis.

Limits on Hypothesis Testing

By themselves, the Monte Carlo simulations will allow us to make some narrow assertions, but for the statistics to have greater local benefit follow-on data gathering and analysis would be needed. In this project we are scaling down national and state level data to a more localized scope. However, that doesn't provide us with testable hypothesis. These statistics could be used as a baseline for understanding the context of other treatments, measurements, or forecasts; but, by themselves, there will be little we can prove or disprove with our simulations. We will be able to make simple assertions about the likelihood and severity of impact certain cybercrime types may have on our sample, but we will not be able to forecast the effectiveness of any local treatments, mitigations, or defensive techniques.

3.1.4 Scope of Work

We chose to limit our risk analysis to the algebraic scaling of IC3 data on Excel spreadsheets to limit the calculations by hand. The data has four main trends (victim count, victim impact, suspect count, suspect impact), but to keep the analysis productive we would proceed all the way through with victim count. This would support an assessment of likelihood. Then we would go through the mathematics again to determine severity by scaling the values of victim impact. With those two categories available, we would have the minimum amount of data needed to conduct a quantitative assessment of likelihood and severity. This would meet the minimum expectations of most risk assessments' data analysis needs.

We would need to scope the data to understand it in a verifiable context. Since our goal is to analyze a project-sized scaling, we would need to couch that target between verifiable guideposts. So, we chose to scale state-level data to our city's population. Alongside this, we could scale state-level data to a small organization's size (e.g. 150 people as stakeholders). Since that scaled data would be calculated, we would want to present it in context with states and territories that represented high and low levels of cybercrime (California and American Samoa). This way we would set a scope that would allow a reader to see the effect of scaling from known states down to the small business organization level.

3.1.5 Variables

Not all of the collected IC3 data categories were stable over time. As the reports were produced year to year, new threats emerged and the cybercrime field changed. This meant that some categories that were present at the older end of our analysis (2016) were not present at the end of our sampling (2022) or later. Because of this,

we had to ignore some categories or combine others. Variability in criminal activity affected reporting.

This had indirect effects on our analysis. Check Fraud and Credit Card Fraud, as activity titles, changed over the years. Threats to people, in the form of both Terrorism and Stalking, changed over the years. This meant that we chose to ignore some categories that would seem to have value to commercial businesses while combining counts into one category in others. Our picking and choosing among the categories genuinely shapes the risk picture we're considering as we do the risk assessment. Our projects and labs can't live in a fictional world where entire threat categories don't exist; but, we had to keep the data calculatable in order to proceed.

3.1.6 Assumptions, Risks, and Limitations

We set out to better understand the rates of computer-related crimes in our area using public data sources. We drew from FBI IC3 annual reports and Chattanooga Police open data records. The IC3 reports provided categorical breakdowns of different types of internet-related crimes. In the past, we were able to find per-state breakdowns of those crimes. The FBI would provide four views of computer crime: overall count, overall financial impact, victim count, and per-victim financial impact. This was available both nationally and per state. When following up in 2026, we noticed we were only able to immediately discover the national-level summaries of crime.

Anecdotally, we know that we have witnessed several of these crimes at rates higher than shown in these tables (Tables 3.2, 3.3, 3.4). We suspect that we're seeing numbers this low because few victims are taking the steps necessary to report computer crimes to federal authorities. For example, when companies or individuals encounter ransomware, their reactions may involve repairing the affected endpoint but not informing law enforcement. Nonetheless, those authorities have been the public's observable source of crime reporting. While we continue to use this data, we may need to accept that true counts of cybercrime in the city must be scaled by an unknown factor.

The Chattanooga Police Department, like other departments of the City of Chattanooga, participate in an open data project. Anonymized crime reports are provided to the public in a manner that lets us track by approximate region of the city and by category of crime. We chose to do this because we recognized some crimes would be reported to local police that were not reported to federal authorities. Some crimes, such as physical break-ins, were more likely to be handled by local police. Since the systems involved are in-scope for physical damage, we felt the Chattanooga Police Department's open data would be the best public source for supporting criminal threat forecasts for events like burglary from a vehicle. Some of the crime categories, such as extortion or threats against a person, could be reported to either federal IC3 or local police; the nature of the threats might be different; but, there may be more direct physical threats reported to local police; there is no formal mimetic analysis that would shuttle a threatening complaint to one law enforcement agency or another. Meanwhile, some locally policed crimes may focus upon: employees, facility

safeguards, loss of computer equipment in public, loss of computer equipment by theft from facilities or vehicles, and so on.

As we examine the data from the public data sources and we feel that the reported rates are less than our experience would justify, we have to recognize that there may be shortcomings in the reported data. Perhaps we would do well to scale the rates up to meet our experience. However, those arbitrary inflations would damage the validity of our academic risk calculations. So, for the time being, we leave the inflation or deflation of the end results for the reader as a localized adjustment. Since our main goal is to show a method for determining the risks through the reported data, it could be argued that the exact numbers are less relevant than the demonstrated methods.

Limitations Imposed by Federal Law Enforcement in Tennessee

During our research, one shocking fact came to light. At the Cybersecurity Summit conference in Nashville, TN in 2023, Derek Rousseau, a Special Agent from the FBI Nashville Cyber Squad, presented and answered a Q&A. To our surprise, the special agent answered a direct question, How much money needs to be at stake for the Federal Bureau of Investigation (FBI) to get involved in a cybercrime? His answer was that the US Attorney in Nashville wanted to see at least \$100,000 in damage for domestic crimes or \$400,000 in damage for international crimes. For damage amounts below that, it was not effective to devote an FBI special agent to investigate the crime. As a result, the special agent encouraged the reporting of smaller crimes so that cases could be aggregated against a suspect to levels that merited federal investigation and prosecution. Later, as we compare IC3 crime data, we'll see that many of the annual totals are below these limits. This suggests that the US is prosecuting only a fraction of those reported crimes that affect the population.

Limitations Derived from Reporting

Another limitation on our projections arises from the voluntary nature of the reporting. Since victims or defenders need to advise the IC3 of a security event, only those who follow through with the voluntary reporting actions will get counted. Commercial defenders will see that in their experience only a small number of daily cybersecurity events will seem worthy to report to federal law enforcement; the scope of true attacks would seem through those anecdotes to be much greater than the reported numbers. Since all of the reports are likely to be initiated by those who speak up, we cannot escape the experimental design problems associated with a convenience sample when using IC3 data for our statistics.

As we've seen and will see, the grouping of those reports does not match our experimental needs. States and nations are simply too large to be of practical benefit to defenders. Most small businesses and organizations will need the value of public information; commercial vendors can provide similar data for a fee; we need the free data, but we need to tailor it down to the size of a defending organization to apply it realistically to risk assessments for cybersecurity defense.

3.1.7 Significance of the Project

We hope to take the mystery out of public statistics for cybersecurity use. We want to provide the types of workshop examples we've seen locally in open data workshops [6] but in technical note form. Ideally, readers should be able to follow these notes and then craft their own projects for forecasting cybersecurity risk using publicly available data sources.

3.2 Literature Review

3.2.1 Overview of Existing Solutions or Approaches

3.2.2 Comparisons of Technologies or Frameworks

3.2.3 Gaps in the Literature

We haven't found many examples outside of college classrooms that allow readers to follow the extrapolation process and formulate their own forecasts for cybersecurity risks. While there are excellent books on the topics of statistics and cybersecurity, we haven't found works that took public data like this and used it for other purposes.

In our hometown of Chattanooga, there have been public non-profit efforts to use open data in workshops that are free and open to the public. Most of the data sets used for those activities are part of local government's participation in an open data initiative. [7] By working through Jupyter notebooks and group discussions, participants might learn more about public bicycle rental use, for example. Our interests, meanwhile, lay firmly within the cybersecurity realm.

3.3 Methodology

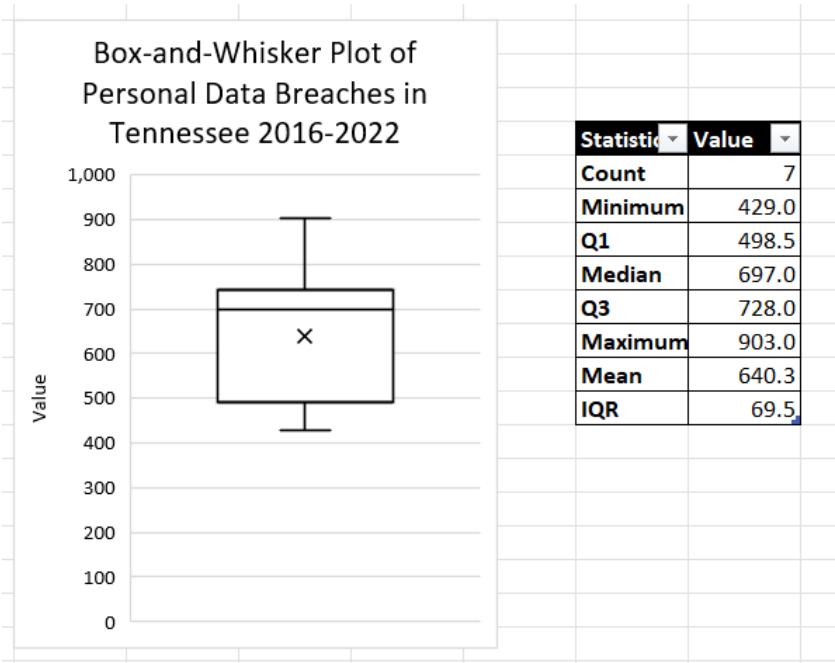


Fig. 3.3: Seven year summary of Personal Data Breaches in Tennessee reported to IC3.

We took cybercrime data from a seven year span of IC3 publications. The annual reports for the nation and for each state showed a count of victims per category of cybercrime. The US FBI defined each category for their report. We put that data into a matrix, organized by year and category of crime.

Each state we chose got its own matrix. We noticed that three states typically had high counts of crime; they were: California, Texas, and Florida. We noticed that US territories, like American Samoa, US Virgin Islands, and Puerto Rico, were reported also as if they were a state. These smaller areas with lower populations also showed low counts of cybercrime. We chose these high crime areas (e.g. California) and low crime areas (e.g. American Samoa) as our guides for choosing maximum and minimum limits. Our desired target areas were chosen by the target's home state; our initial runs included Chattanooga, Tennessee.

3.3.1 Using State Data to Forecast a Company's Data

Our main data sets are provided in the public interest; they're based on large units of people like states and nations. As system operators, we have a much smaller interest; we're interested in company sized elements. Instead of being concerned

with the cybercrime fate of over 300 million Americans or 9 million Tennesseans, we are probably more interested in what might happen to 100 stakeholders.

To smooth out the counts of cybercrime into rates, we adjusted these state victim counts per capita. We chose the US Census reports for those years to be the population values of the states. This would provide us with a per capita rate of cybercrime based on the chosen state which represented the chosen limit or target in the distribution.

Formally, we note this process below as shown in Figure 3.4. R represents our result. N_t represents our constant for target sample population and is used for multiplying the rate to scale it to match the population of the target. t represents our starting year for the sample from IC3 and coordinates the rows until it progresses to the final year chosen which is denoted by n . i represents our chosen category index of cybercrime. j represents our year index and is built with t until it progresses to the final year of the sample denoted by n . P_j represents the population of the target city at year j , drawn from matrix P .

$$R_j = N_t \sum_{j=t}^{t+6} \frac{X_{j,i}}{P_j}, \quad \text{where } X_{j,i} = \begin{cases} M_{j,i} & \text{if victim count} \\ V_{j,i} & \text{if victim impact} \\ S_{j,i} & \text{if suspect count} \\ D_{j,i} & \text{if suspect impact} \end{cases}$$

Fig. 3.4: Notation for the statistical calculation of the security event matrix adjusted for per-capita rates of incidence.

Five matrices inform this calculation. The first, P as shown in Figure 3.5, records the target city population for each year of the sample as a single column.

$$P = \text{Years} \left\{ \begin{array}{c} t \\ \vdots \\ j \\ \vdots \\ n \end{array} \right\} \left[\begin{array}{c} \text{Population} \\ \hline p_t \\ \vdots \\ p_j \\ \vdots \\ p_n \end{array} \right]$$

Fig. 3.5: Structure of the sample population matrix indexed by year.

The second matrix, M as shown in Figure 3.6, organises the raw cybercrime counts from IC3 by category and year, where each entry $a_{j,i}$ is a count that will be adjusted to a per-capita rate.

$$M = \text{Years} \left\{ \begin{array}{c} t \\ \vdots \\ j \\ \vdots \\ n \end{array} \right. \left[\begin{array}{c} \overbrace{\begin{array}{ccccccc} a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{array}}^{\text{Victim Counts by Categories}} \end{array} \right]$$

Fig. 3.6: Structure of the security event matrix for victim counts by cybercrime categories indexed by year.

The third matrix, V as shown in Figure 3.7, organises the victim impact in USD from IC3 by category and year, where each entry $a_{j,i}$ is a count that will be adjusted later to a per-capita rate.

$$V = \text{Years} \left\{ \begin{array}{c} t \\ \vdots \\ j \\ \vdots \\ n \end{array} \right. \left[\begin{array}{c} \overbrace{\begin{array}{ccccccc} a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{array}}^{\text{Victim Impacts by Categories}} \end{array} \right]$$

Fig. 3.7: Structure of the security event matrix for victim impacts, as in dollars of damage, by cybercrime categories indexed by year.

The fourth matrix, S as shown in Figure 3.8, organises the count of suspects from IC3 by category and year. Each entry $a_{j,i}$ helps us to understand how many suspects are operating within an area of concern. This is a count that will be adjusted later to a per-capita rate.

$$S = \text{Years} \left\{ \begin{array}{c} t \\ \vdots \\ j \\ \vdots \\ n \end{array} \right\} \left[\begin{array}{ccccccc} & \overbrace{\text{Suspect Counts by Categories}} & & & & & \\ a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{array} \right]$$

Fig. 3.8: Structure of the security event matrix for suspect counts by cybercrime categories indexed by year. These counts consider the number of actors who engage victims in a given area.

The fifth matrix, D as shown in Figure 3.9, organises the damages attributed by suspects in USD from IC3 by category and year, where each entry $a_{j,i}$ is a count that will be adjusted later to a per-capita rate. This helps us to understand rates of effective impact, per suspect, in a given area.

$$D = \text{Years} \left\{ \begin{array}{c} t \\ \vdots \\ j \\ \vdots \\ n \end{array} \right\} \left[\begin{array}{ccccccc} & \overbrace{\text{Suspect Damage by Categories}} & & & & & \\ a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{array} \right]$$

Fig. 3.9: Structure of the security event matrix for suspect damage, in dollars by perpetrators, by cybercrime categories indexed by year. This tally helps us understand how damaging each actor is to victims.

We apply these matrices to the formula shown in Figure 3.4. The entries $a_{j,i}$ are the raw counts taken from the IC3 state-level reports for the chosen source state. Dividing each entry by P_j converts it to a per-capita rate; multiplying by N_t then scales that rate to match the population of the target city. The population values in P provide the year-by-year context for N_t and confirm the representativeness of the scaling. Summing the selected column i of the per-capita-adjusted M across all years yields the final result R .

Table 3.1: Changes to population in Key Areas

Year	American Samoa Population	References	City of Chattanooga Population	References	State of Tennessee Population	References	State of California Population	References
2016	52,245	[8]	177,746	[9]	6,646,010	[10]	39,149,186	[11]
2017	51,586	[12]	179,530	[13]	6,708,799	[14]	39,337,785	[15]
2018	50,908	[16]	181,918	[17]	6,771,631	[18]	39,437,463	[19]
2019	50,209	[20]	182,799	[21]	6,829,174	[22]	39,437,610	[23]
2020	49,751	[24]	181,560	[25]	6,925,619	[26]	39,538,223	[27]
2021	49,225	[28]	181,163	[29]	6,968,351	[30]	39,152,927	[31]
2022	48,342	[32]	184,086	[33]	7,051,339	[34]	39,125,347	[35]

City population based on the US Census reported population for the City of Chattanooga itself. Surrounding satellite cities and unincorporated areas within the Hamilton County, Tennessee, were not included. Population for the State of Tennessee provided perspective on calculated rates in later tables. Population for American Samoa and the State of California were included for perspective on maximum and minimum counts in later tables.

Corporate and Personal Data Breaches

Table 3.2: Projected In-City Victim Counts, City of Chattanooga, Threats to System Intrusion

Year	Data Breach	Personal Data Breach	
2016	1	11	[36]
2017	1	13	[37]
2018	0	18	[38]
2019	0	13	[39]
2020	0	18	[40]
2021	0	19	[41]
2022	1	23	[42]

Counts deduced from IC3 Victim Complaints for Tennessee adjusted for a per-capita rate and then multiplied by the city’s population for that year.

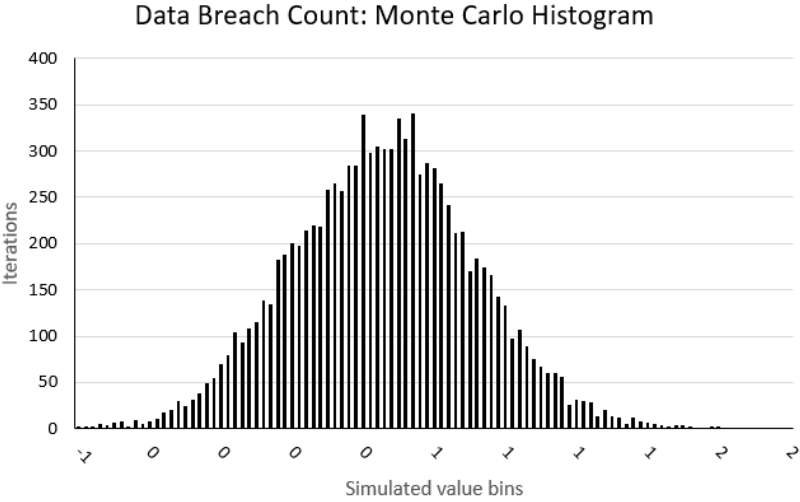


Fig. 3.10: We ran a Monte Carlo simulation of 10,000 iterations to spread random instances between confidence intervals taken from means of a seven year sample. We bounded between 0 and 2 possible data breaches over a year.[43] [44]

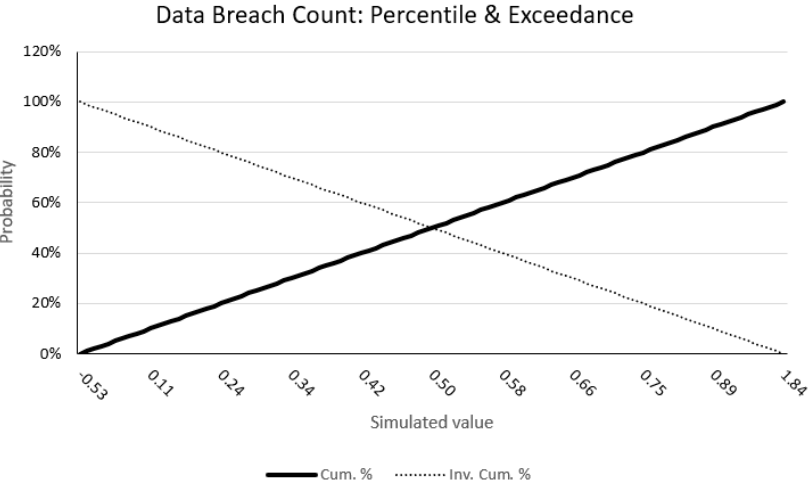


Fig. 3.11: A Monte Carlo simulation suggests that there is a 50% probability that 0.5 or more data breaches will occur within the sample. This unusual result is because the confidence bounds are between 0 and 1. We interpret this as a 50% probability that one data breach will occur. [45] [46]

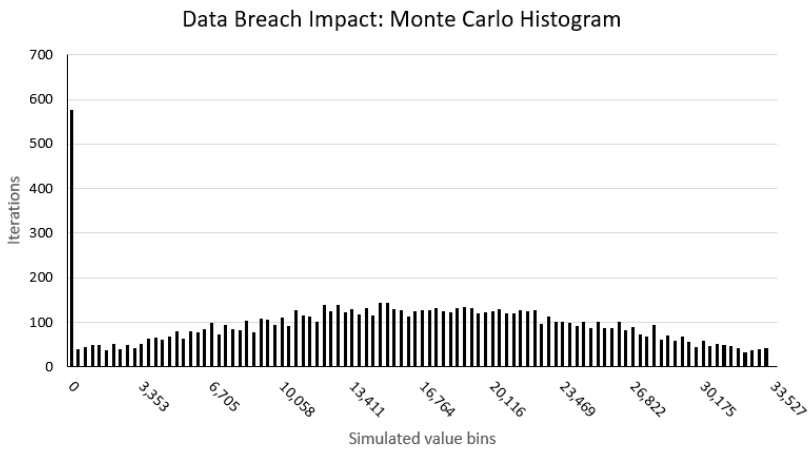


Fig. 3.12: We ran a Monte Carlo simulation of 10,000 iterations to spread random instances between confidence intervals taken from means of a seven year sample. We set a lower bound that the breach would cost at \$0. We set an upper bound that the breach would cost \$33,527.[47] [48]

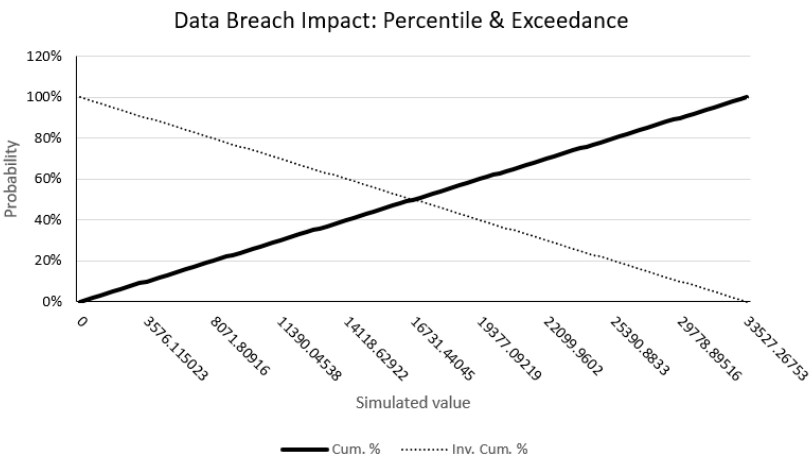


Fig. 3.13: A Monte Carlo simulation suggests that there is a 90% probability that data breaches will cost \$3,576 or more in a year. There is a 50% likelihood that they will cost \$16,731 or more in a year. There is a 10% chance that they will cost \$29,778 or more in a year. [49] [50]

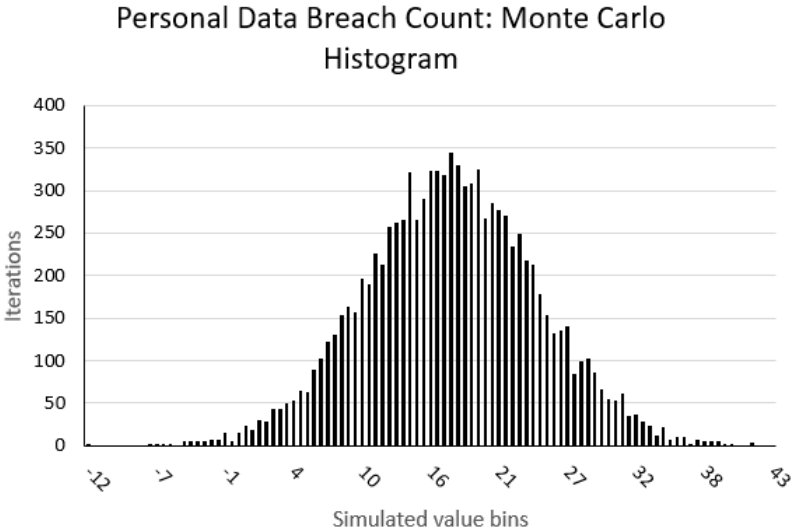


Fig. 3.14: We ran a Monte Carlo simulation of 10,000 iterations to spread random instances between confidence intervals taken from means of a seven year sample. We bounded between 6 and 29 possible personal data breaches over a year.[51] [52]

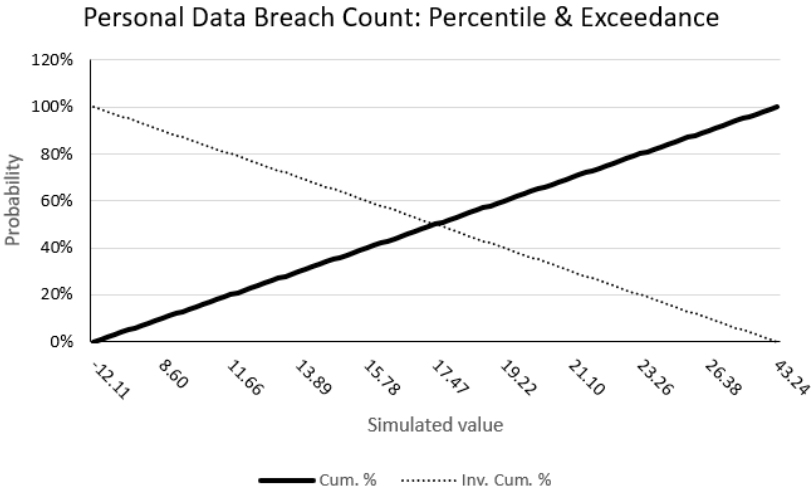


Fig. 3.15: A Monte Carlo simulation suggests that there is a 10% probability that at least 8 personal data breaches will occur within the sample. There is a 50% chance that 17 or more data breaches will occur within the sample. There is a 90% chance that 26 personal data breaches [53] [54]

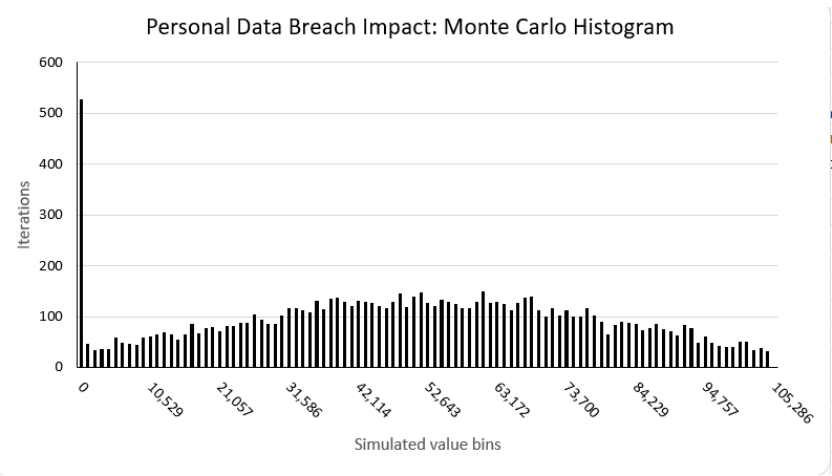


Fig. 3.16: We ran a Monte Carlo simulation of 10,000 iterations to spread random instances between confidence intervals taken from means of a seven year sample. We set a lower bound that the breaches would cost \$0. We set an upper bound that the breaches would cost \$105,286.17.[55] [56]

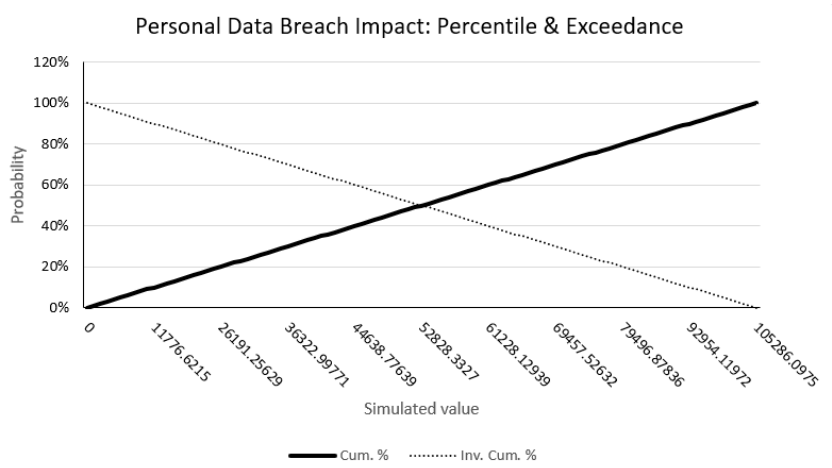


Fig. 3.17: A Monte Carlo simulation suggests that there is a 90% probability that personal data breaches will cost \$11,776 or more in a year. There is a 50% likelihood that they will cost \$52,828 or more in a year. There is a 10% chance that they will cost \$92,954 or more in a year. [57] [58]

Threats to System Intrusion

Table 3.3: Projected In-City Victim Counts, City of Chattanooga, Threats to System Intrusion

Year	Malware	Phishing	Ransomware	
2016	1	7	1	[59]
2017	1	9	0	[60]
2018	0	8	0	[61]
2019	1	8	0	[62]
2020	0	9	0	[63]
2021	0	11	1	[64]
2022	0	7	0	[65]

Counts deduced from IC3 Victim Complaints for Tennessee adjusted for a per-capita rate and then multiplied by the city’s population for that year.

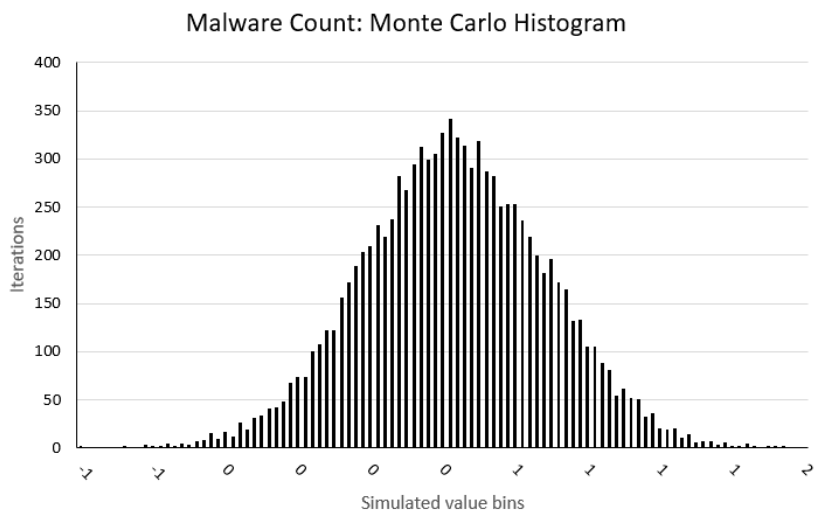


Fig. 3.18: We ran a Monte Carlo simulation of 10,000 iterations to spread random instances between confidence intervals taken from means of a seven year sample. We bounded between 0 and 2 possible malware events over a year.[66] [67]

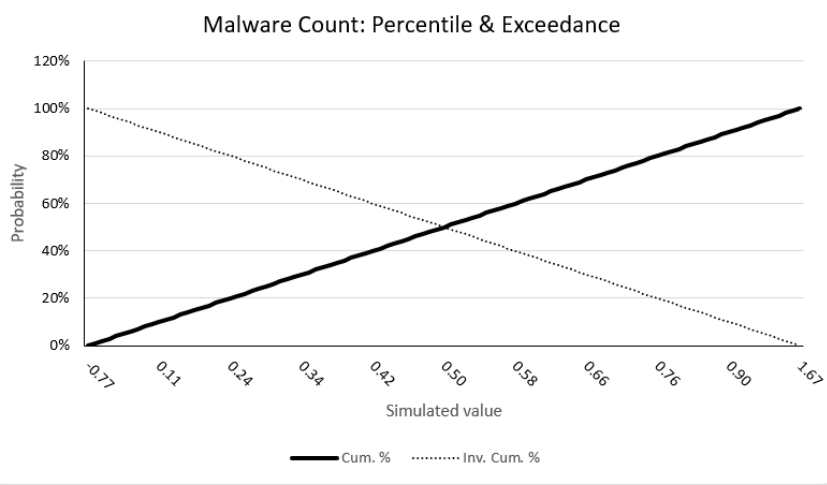


Fig. 3.19: A Monte Carlo simulation suggests that there is a 50% probability that 0.5 or more malware events will occur within the sample. This unusual result is because the confidence bounds are between 0 and 1. We interpret this as a 50% probability that one malware event will occur. [68] [69]

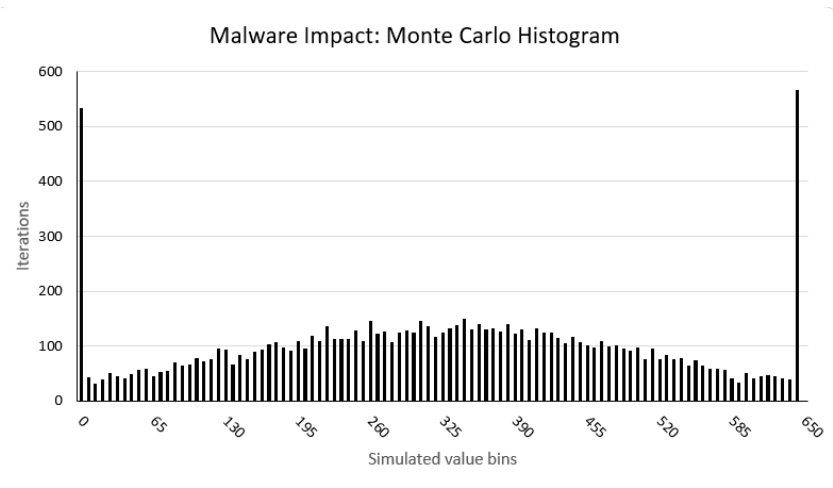


Fig. 3.20: We ran a Monte Carlo simulation of 10,000 iterations to spread random instances between confidence intervals taken from means of a seven year sample. We set a lower bound that the events would cost at \$0. We set an upper bound that the events would cost \$649.99.[70] [71]

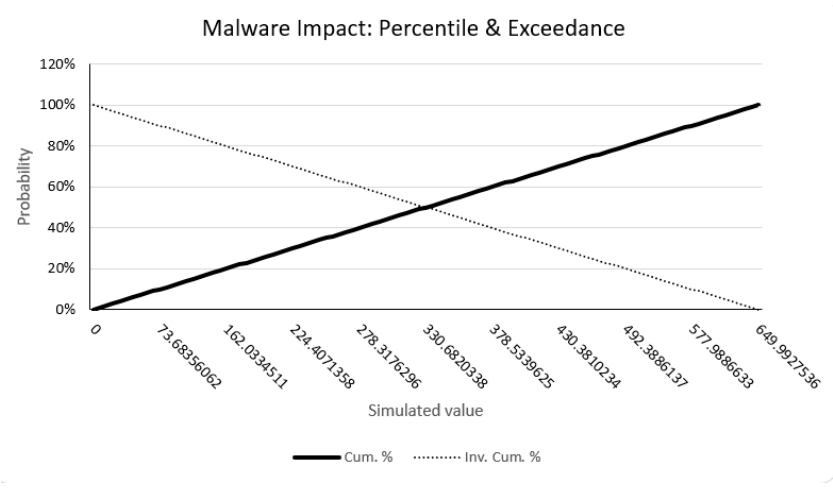


Fig. 3.21: A Monte Carlo simulation suggests that there is a 90% probability that malware events will cost \$73.68 or more in a year. There is a 50% likelihood that they will cost \$330.68 or more in a year. There is a 10% chance that they will cost \$577.98 or more in a year. [72] [73]

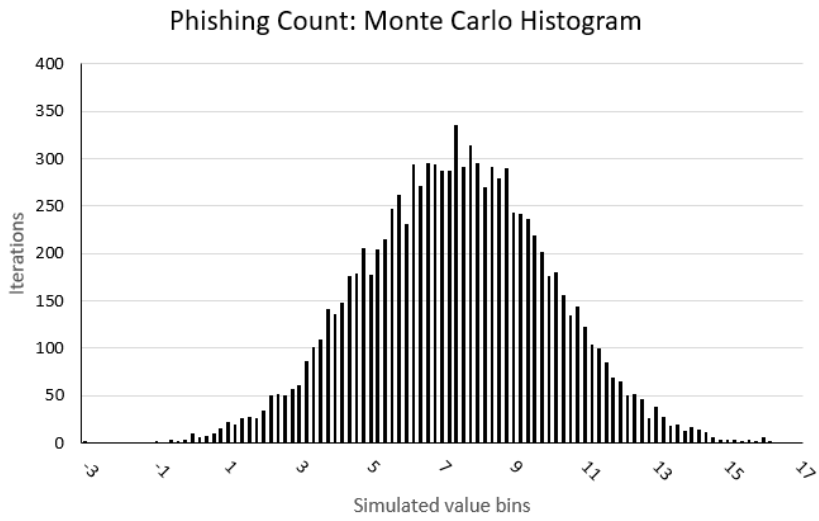


Fig. 3.22: We ran a Monte Carlo simulation of 10,000 iterations to spread random instances between confidence intervals taken from means of a seven year sample. We bounded between 3 and 12 possible phishing events over a year.[74] [75]

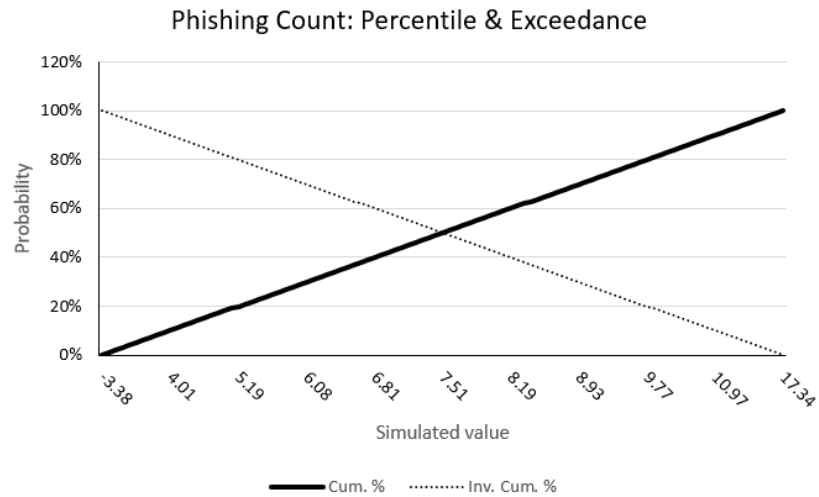


Fig. 3.23: A Monte Carlo simulation suggests that there is a 90% probability that 4 or more phishing events will occur within the sample. There is a 50% probability that at least 7 events will occur. There is a 10% probability that 10 or more events will occur. [76] [77]

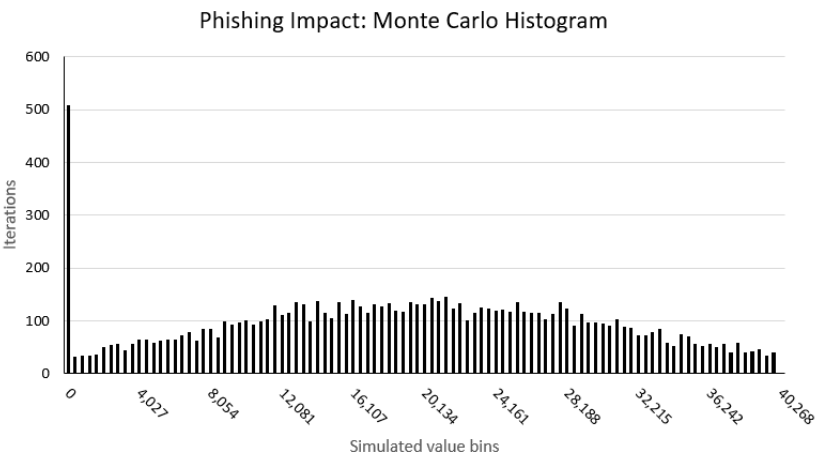


Fig. 3.24: We ran a Monte Carlo simulation of 10,000 iterations to spread random instances between confidence intervals taken from means of a seven year sample. We set a lower bound that the events would cost at \$0. We set an upper bound that the events would cost \$40,268.[78] [79]

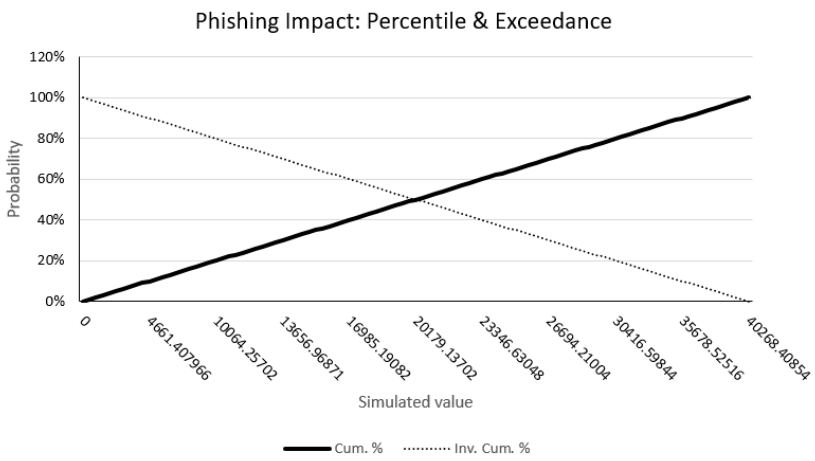


Fig. 3.25: A Monte Carlo simulation suggests that there is a 90% probability that phishing events will cost \$4,661.40 or more in a year. There is a 50% likelihood that they will cost \$20,179.13 or more in a year. There is a 10% chance that they will cost \$35,678.52 or more in a year. [80] [81]

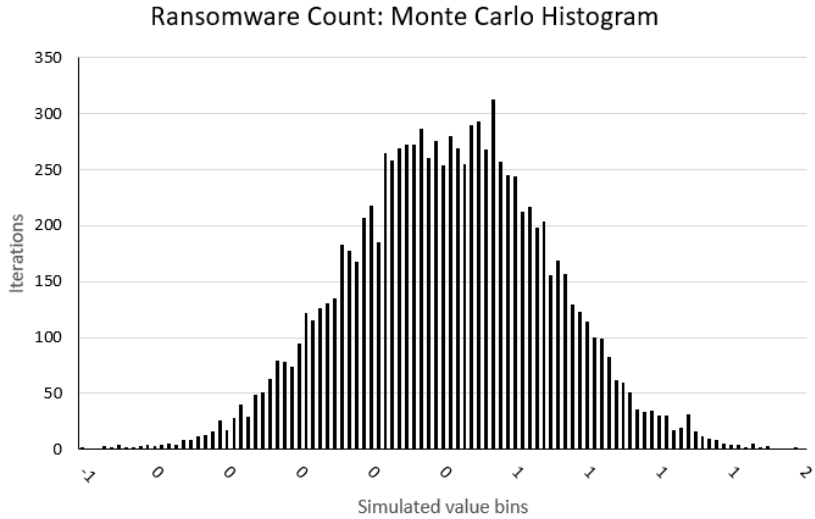


Fig. 3.26: We ran a Monte Carlo simulation of 10,000 iterations to spread random instances between confidence intervals taken from means of a seven year sample. We bounded between 0 and 1 possible ransomware events over a year.[82] [83]

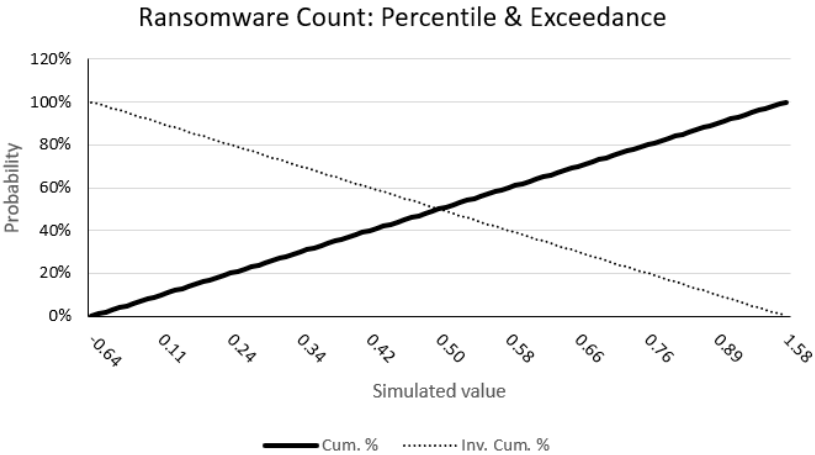


Fig. 3.27: A Monte Carlo simulation suggests that there is a 50% probability that 0.5 or more ransomware events will occur within the sample. This unusual result is because the confidence bounds are between 0 and 1. We interpret this as a 50% probability that one malware event will occur. [84] [85]

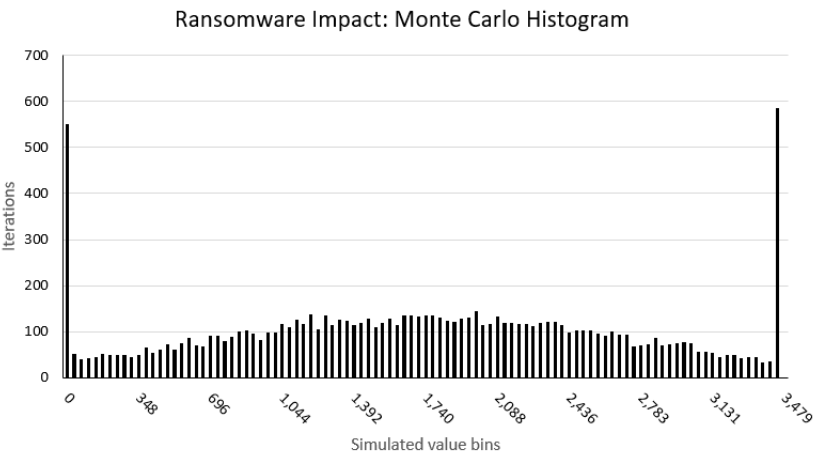


Fig. 3.28: We ran a Monte Carlo simulation of 10,000 iterations to spread random instances between confidence intervals taken from means of a seven year sample. We set a lower bound that the event would cost at \$0. We set an upper bound that the event would cost \$3,479.[86] [87]

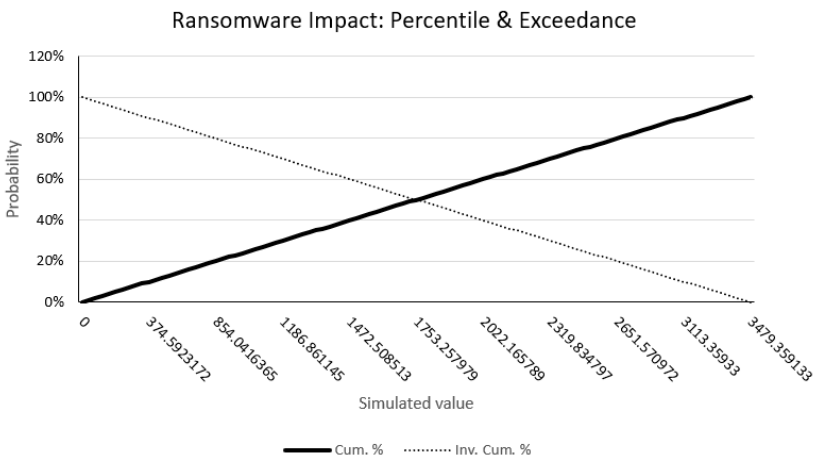


Fig. 3.29: A Monte Carlo simulation suggests that there is a 90% probability that ransomware events will cost \$374.59 or more in a year. There is a 50% likelihood that they will cost \$1,753.25 or more in a year. There is a 10% chance that they will cost \$3,113.35 or more in a year. [88] [89]

Threats to Personnel

Table 3.4: Projected In-City Victim Counts, City of Chattanooga, Threats to Personnel

Year	Crimes Against Children	Extortion	Threats, Stalking, Harassment, and Terrorism	
2016	0	9	7	[90]
2017	0	7	6	[91]
2018	0	20	7	[92]
2019	0	19	5	[93]
2020	1	31	6	[94]
2021	0	0	4	[95]
2022	1	18	0	[96]

Counts deduced from IC3 Victim Complaints for Tennessee adjusted for a per-capita rate and then multiplied by the city’s population for that year. Threats, stalking, and harrassment counts were combined with terrorism because the mechanisms were similar.

Fig. 3.30: The Monte Carlo simulation determined that there would be no reasonable chance of crimes against children in this sample. [97] [98]

Fig. 3.31: Since the Monte Carlo simulation determined that there would be no reasonable chance of crimes against children in this sample, we can see that there would be no financial impact. [99] [100]

Threats to Theft of Data Via Network

Table 3.5: Projected In-City Victim Counts, City of Chattanooga, Theft of Data Via Network

Year	Intellectual Property Rights (IPR) and Counterfeit	Identify Theft	Business Email Compromise (BEC)	
2016	1	7	4	[101]
2017	1	8	4	[102]
2018	0	10	7	[103]
2019	1	7	8	[104]
2020	1	10	6	[105]
2021	1	13	8	[106]
2022	0	14	10	[107]

Counts deduced from IC3 Victim Complaints for Tennessee adjusted for a per-capita rate and then multiplied by the city’s population for that year.

3.3.2 Methodology for Monte Carlo Simulations in Excel

We chose IC3 crime data from American Samoa to act as our 10% confidence limit. We chose data from Californiz to act as our 90% confidence limit. We set those limits by looking at seven year spans of crime data and taking the median. This means we would be likely to have observations above and below the limits, but that the limits themselves would have a basis in observation. We input and processed the data as follows:

- We imported the data fromIC3 crime tables
- We calculated counts and costs for Chattanooga, American Samoa, and California.
- We imported a Uniform distribution table from the Monte Carlo Toolkit.
- We set median summary counts from American Samoa as our 10% interval.
- We set median summary counts from California as our 90% interval.
- We accepted the Monte Carlo percentages and exceedence curves.

3.4 Analysis

3.4.1 Using Nation or State Data as a Control for Smaller Forecasts

If we apply unadjusted for sample size counts limited to a one-year basis, we could use a Poisson distribution as shown in Figure 3.32.

$$X_{j,i} = \begin{cases} M_{j,i} & \text{if victim count} \\ V_{j,i} & \text{if victim impact} \\ S_{j,i} & \text{if suspect count} \\ D_{j,i} & \text{if suspect impact} \end{cases}$$

$$Y_i = \sum_{j=t}^{t+6} X_{j,i}$$

$$Y_i \sim \text{Poisson}(\lambda_i E_i)$$

$$R_i = \frac{Y_i}{E_i}$$

Fig. 3.32: Summary of applying Poisson distribution to counts from nation and state data.

This would allow us to feed per-state or national data into the formula and better understand the relationship of our projections to a larger view. It could help us to understand where our projected sample's forecast would stand against the whole state overall or nationally. This way, we could use the state and national data as controls for understanding the context of our forecast.

$$r_{j,i} = \frac{Y_{j,i}}{P_j}$$

Fig. 3.33: Empirical rate per year

$$Y_i = \sum_{j=t}^{t+6} Y_{j,i} \quad E_i = \sum_{j=t}^{t+6} P_j$$

Fig. 3.34: Aggregated count and exposure over historical window

$$\bar{r}_i = \frac{Y_i}{E_i} = \frac{\sum_{j=t}^{t+6} Y_{j,i}}{\sum_{j=t}^{t+6} P_j}$$

Fig. 3.35: Directly standardized (weighted mean) rate

$$Y_i \sim \text{Poisson}(\bar{r}_i \cdot E_i)$$

Fig. 3.36: Poisson model for the aggregated count

$$\bar{r}_i \pm 1.96 \sqrt{\frac{\bar{r}_i}{E_i}}$$

Fig. 3.37: Approximate 95% confidence interval for the rate

$$\hat{R}_i \in \left[\bar{r}_i - 1.96 \sqrt{\frac{\bar{r}_i}{E_i}}, \bar{r}_i + 1.96 \sqrt{\frac{\bar{r}_i}{E_i}} \right]$$

Fig. 3.38: Forecast evaluation: is the forecast within the control band?

3.5 Discussion

3.5.1 Why Some Categories Might Be Lower

As we look over some of our generated statistics, we can see that sometimes our expectations were defied. Why, for example, are some of the categories so low in likelihood? At our level of access to the data from IC3, we don't know more of the background details that might point directly to a cause for that. Perhaps law enforcement gets better insight; but, we have to make do with what we have. Let's consider some of the categories and suppose, arbitrarily and based on our experience, why some of the figures might be the way they are.

First, we have the overall effect of the convenience sample. As we mentioned earlier, these reports are based on voluntary disclosures to US FBI IC3. That means

that we could see lower numbers in areas where victims were not motivated to disclose.

Potential victims could also have successful mitigation efforts in place. For example, we see lower corporate data breaches than personal data breaches. Presumably, corporations have security departments that can help them to operate enterprise grade defensive tools when individual people cannot. Also, we see low counts in malware and ransomware when those categories are often in the news. It could be that conventional antivirus scanners and endpoint defenses do a good job of neutralizing known malware threats. If a large number of antivirus users are being protected successfully, then we might have a small number of victimization claims to the IC3.

3.5.2 Crimes Against Children and Complaints

3.5.3 Business Email Compromise Versus Ransomware

A long standing pattern of expectation is that a user will be phished, malware will be deployed on their machine, and then a company will face a ransomware attack. Besides this, we see a business email compromise category. If BEC is a precedent attack, then why aren't its numbers higher?

One of the reasons could be the ramifications of the attacks and their flow into later reporting classifications. BEC attacks could also flow in other directions like credential stuffing attacks or passive surveillance. In credential stuffing, the known business email address might be reused in later follow-on attacks to other systems. If an attacker chooses a victim that's a key individual in the company, then passive surveillance of the email account might yield valuable information. Several 10K reports cite top executives getting their accounts monitored through what looks like a combination of passive surveillance and BEC.

These hint in the numbers can suggest to us an illustration that the overall classification of an attack may not offer direct insight on all of the tactics and techniques that concern us as defenders. This shouldn't surprise us; we see plenty of evidence in MITRE ATT&CK, for instance, that most Advanced Persistent Threat (APT)s use several layers of Tactics, Techniques, and Procedures (TTPs).

Endnotes

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- [9] <https://www.census.gov/quickfacts/chattanoogaacitytennessee>
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- [15] <https://fred.stlouisfed.org/series/CAPOP>
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- [37] original: <https://www.ic3.gov/Media/PDF/AnnualReport/2017State/StateReport.aspx?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2017_IC3Report.pdf
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[43] Sample size was set equivalent to the City of Chattanooga to determine the per capita rates for each year over data from American Samoa and California. The medians of those rates were used to set 10% and 90% confidence intervals with American Samoa and California representing low and high cybercrime counts respectively.

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[93]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2019State/StateReport.aspx?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2019_IC3Report.pdf

[94]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2020State/StateReport.aspx?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2020_IC3Report.pdf

[95]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2021State/StateReport.aspx?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2021_IC3Report.pdf

[96]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2022State/StateReport.aspx?s=47>

tateReport.aspx#?s=47 current: https://www.ic3.gov/AnnualReport/Reports/2022_IC3Report.pdf

[97] sample size was set equivalent to the City of Chattanooga to determine the per capita rates for each year over data from American Samoa and California. The medians of those rates were used to set 10% and 90% confidence intervals with American Samoa and California representing low and high cybercrime impact values respectively.

[98]Monte Carlo Modeling Toolkit authored by Derek E. Brink, CISSP, www.linkedin.com/in/derekbrink. March 2023.

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[101]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2016State/StateReport.aspx#?s=47> current https://www.ic3.gov/AnnualReport/Reports/2016_IC3Report.pdf

[102]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2017State/StateReport.aspx#?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2017_IC3Report.pdf

[103]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2018State/StateReport.aspx#?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2018_IC3Report.pdf

[104]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2019State/StateReport.aspx#?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2019_IC3Report.pdf

[105]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2020State/StateReport.aspx#?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2020_IC3Report.pdf

[106]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2021State/StateReport.aspx#?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2021_IC3Report.pdf

[107]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2022State/StateReport.aspx#?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2022_IC3Report.pdf

